Class 4: Electrical Conductivity

In this class we will first examine why it is of interest to focus our learning process on electrical conductivity. We will recognize that there are different types of charge carrying or conducting species and that these species may exhibit different behaviors, and finally we will understand how conductivity is measured.

If we look at the various gadgets we routinely use these days, it is easy to recognize that several technologies depend on, or make use of electronic properties of materials. Toys, household appliances, and automobiles are examples where several mechanical systems have been replaced or augmented by electronic systems. Whereas previously it was possible to open and repair toys, because they were based on mechanical systems, opening present day toys brings you face to face with an electronic chip and the repair process usually stops there. Most modern automobiles have sophisticated electronics that help them maximize fuel efficiency, or manage the braking process, or control the traction of the wheels etc. The above are just a few examples of how pervasive electronics has become in our present day society.

Of the electronic properties that a material can display, conductivity is of particular interest for us for a few different reasons. Firstly, conductivity is possibly the 'most commonly used' electronic property in a technological sense, both in terms of good conductors for wires and bad conductors to provide insulation for the wires. Secondly, it turns out that best conductors of electricity and the worst conductors of electricity, vary in conductivity by over 24 orders of magnitude. For example Silver has a conductivity of approximately 10^{7} $\Omega^{-1}m^{-1}$, whereas Teflon has a conductivity of less than 10^{-17} $\Omega^{-1}m^{-1}$. There is almost no other property that displays such a significant variation in its manifestation in various materials. Therefore from a technological perspective it is of interest to study electrical conductivity, due to its widespread use, and from a scientific perspective it is of interest to see if it is possible to identify the theories that can explain such a large variation in the property.

As we take a closer look at some of the aspects associated with conductivity, it is important to recognize that in the most general sense, electrical conductivity is the transport of charge. The charge that is being transported can be carried by different species, or charge carriers. Different charge carriers may have different mass, interact with their surroundings differently, may respond to changes in external environments differently, and may face different limitations in terms what they can do and cannot do.

The charge carriers most commonly encountered in physics and engineering are:
a) Electrons
b) Holes
c) Ions

In a single circuit, different sections of the circuit may have different charge carriers. For example, consider a circuit that consists of an electrochemical power sources connected to an external load, as shown in Figure 4.1

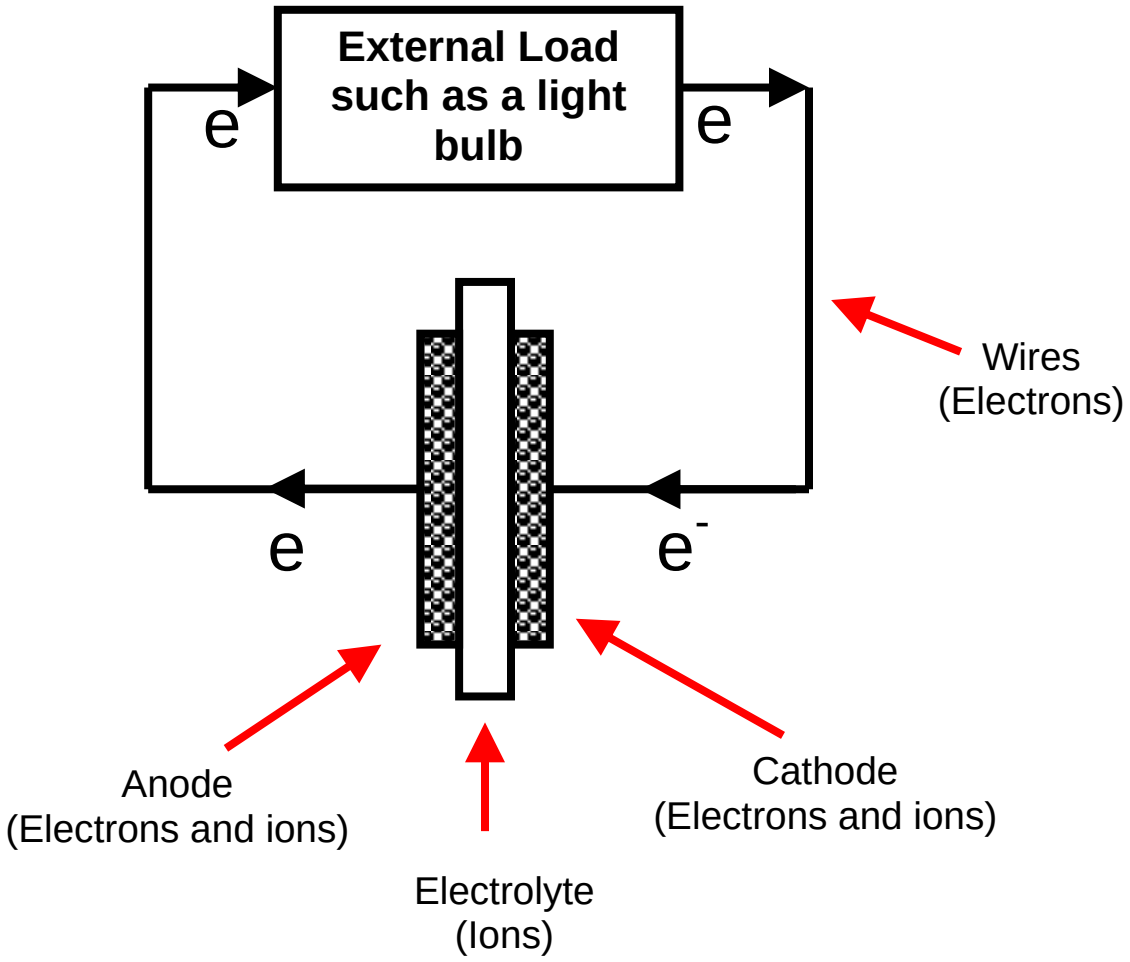


Figure 4.1: A schematic that shows the charge carriers (indicated within brackets) in various sections of a circuit which connects an electrochemical power source, such as a battery, to an external load such as a light bulb. The anode, electrolyte, and the cathode, together constitute the electrochemical power source

In this circuit, the wires connected to the external load have electrons as the charge carriers; in the electrolyte, ions are charge carriers; and in the electrodes (anode as well as cathode), both electrons as well as ions are charge carriers. This type of circuit is quite common place – virtually every battery operated gadget is an example of the above.

Consider also a circuit that connects to a p-n-p transistor, as shown in Figure 4.2:

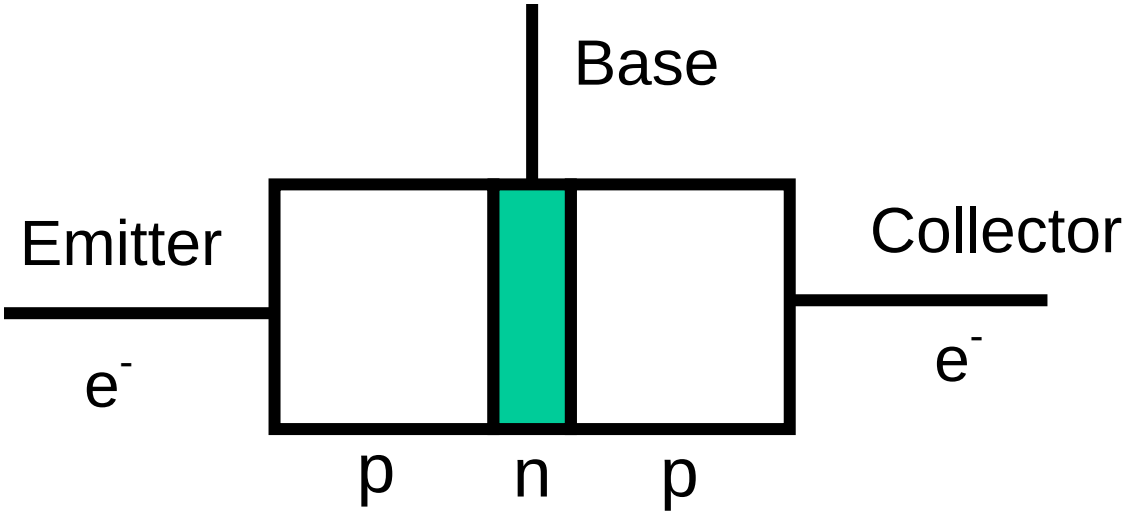


Figure 4.2: A schematic that shows a p-n-p transistor. The charge carriers, from left to right, are electrons, holes, electrons, holes, and electrons

The charge carriers in the wires that connect to the device are electrons. Within the device, the charge carriers are holes in the regions that show p-type semiconductivity, and are electrons in the regions that show n-type semiconductivity. In particular, in Figure 4.2 above, going from left to right, the charge carriers are electrons, holes, electrons, holes, and electrons. Electronic circuits in most modern devices are built using several transistors and hence this example is also very commonly encountered.

The two examples above highlight the fact that we are now routinely using devices, gadgets and technologies that have multiple charge carriers, it is just that we are often not aware of this information, we simply think of 'current' as being associated with 'electrons'.

As indicated earlier, in the most general sense, electrical conductivity is the transport of charge. Now that we recognize there can be different types of charge carriers, it is of interest to see how conductivity can be measured and how such measurement handles the possibility of different types of charge carriers.

Direct Current (DC) Conductivity measurement:

This is the form of electrical conductivity measurement that most of us are familiar with. However, even in this form of conductivity measurement, there are two variations possible: **Two probe measurement**, and **four probe measurement**. As the names suggest, in the two probe measurement, the sample is simultaneously contacted in two places and conductivity is measured, whereas in the four probe measurement, the sample is simultaneously contacted at four places to measure its conductivity. While the difference in these two methods may seem trivial at first glance, there is a specific issue with the two probe measurement which is effectively addressed by the four probe

measurement, and hence the later is now the more commonly accepted technique for DC conductivity measurement.

In a DC conductivity measurement process, in principle current from a DC source should flow through the sample and the potential difference that develops across the sample must be measured. Using Ohm's law the resistance of the sample can be determined, and then using the dimensions of the sample, the conductivity of the sample can be determined.

Ohm's law can be written as:

$$V = IR$$

Where 'V' is the potential difference across the sample, when the current 'I' is flow through it.

Once 'R' is determined, the resistivity of the sample can be determined using the relationship:

$$R = \rho l/A$$

Where 'ρ' is the resistivity of the sample, 'l' is the length of the sample, and A is the cross sectional area of the sample. We will assume that the length and cross sectional area of the sample are uniform and are therefore each of a single value.

The conductivity of the sample is then simply the inverse of the resistivity.

Conductivity 'σ' is given by:

$$\sigma = 1/\rho$$

The units for conductivity are $\Omega^{-1}m^{-1}$

Experimentally, the challenge therefore is to measure the V and I values accurately and correctly, and also to determine the length and cross sectional area of the sample.

The two probe and four probe methods differ in how well they determine the 'V' value that is of relevance. As it turns out, the two probe measurement results in a higher value for the 'V' than is actually the case, and hence it over estimates the resistance of the sample, and therefore underestimates the conductivity of the sample. The four probe method enables the measurement of 'V' in a significantly more accurate manner, and hence is the preferred method for DC conductivity measurement.

In Figure 4.3, a schematic is shown of the two probe measurement technique to obtain conductivity of the sample.

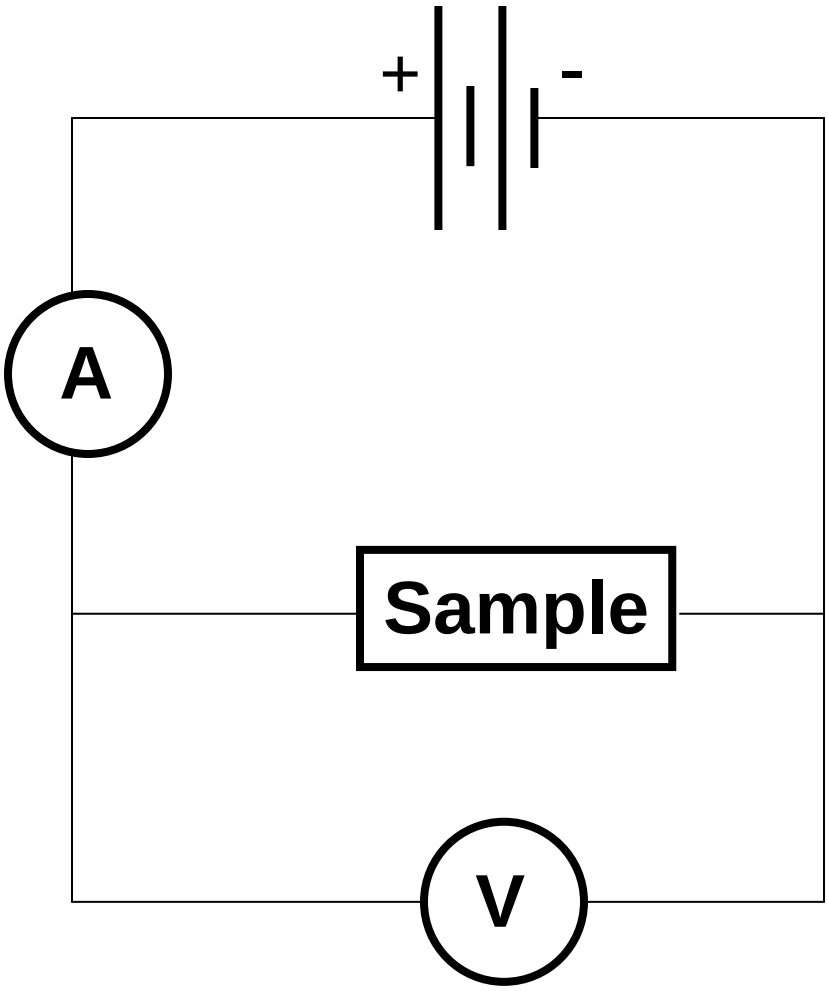


Figure 4.3: A two probe measurement of the electrical conductivity of a sample. A battery serves as the source of DC power, an ammeter measures the current in the circuit, and a voltmeter measures the potential drop across the sample.

In the two probe measurement of electrical conductivity of the sample, a battery may serve as the source of DC power, an ammeter can be used to measure the current in the circuit, and a voltmeter can be used to measure the potential drop across the sample.

When any two surfaces come in physical contact with each other, the contact is never perfect, nor are the surfaces themselves perfect. Typically surfaces have thin non conducting oxide layers on them, and may very well be rough at an atomic level even if polished and cleaned at a macroscopic level. As a result, when two wires come in contact with each other or when a wire is attached to any electrical component or device, the current experiences a significant resistance as it tries to flow from one contacting surface to the next. This resistance is simply referred to as 'Contact Resistance', which we shall denote as R_c. . In addition, the wires used to connect to the sample also have some finite resistance. If we ignore the resistance of the wires, we still find that the manner in which the voltmeter is connected in the two probe measurement technique, results in a situation where the potential drop measured by it will be the result of the potential drop across the

sample, as well as the potential drop caused by the contact resistances present where the wires from the external circuit contact the sample on either side of the sample.

In other words, if the resistance of the sample is 'R_s', and the current in the circuit is 'I', the potential drop measured by the voltmeter is given by:

$$V = IR_c + IR_s + IR_c$$

The contact resistance term appears twice, because contacts have to be made on either side of the sample. The potential as a function of position will therefore be as shown in Figure 4.4 below:

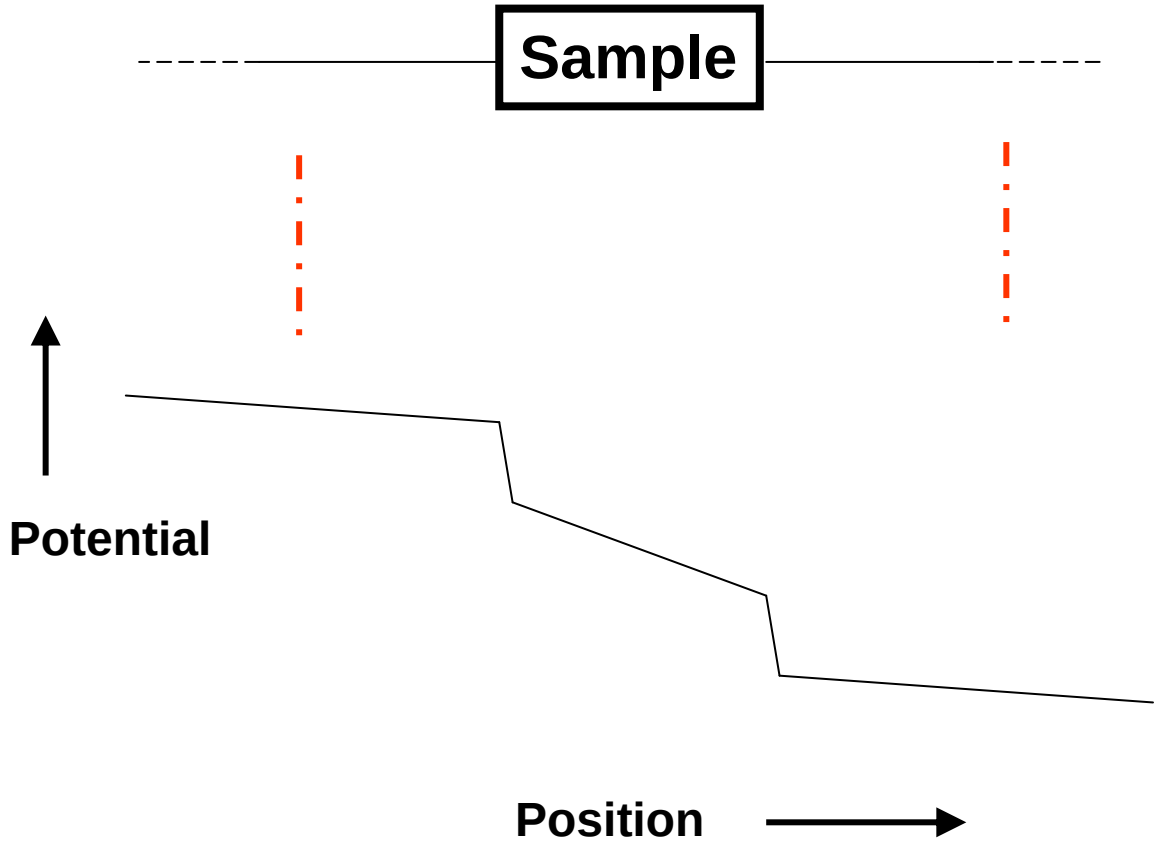


Figure 4.4: Potential as a function of position. The red dotted lines indicate the positions at which a two probe measurement technique will measure the potential drop across the sample.

Since we have mentioned that R_c can be a significant quantity, it is now evident from the equation as well as the figure above, that we will be over estimating the value of V.

The measurement of the potential drop attributed to the sample will be much more accurate if the positions at which the potentials are measured, are changed, as indicated in Figure 4.5 below.

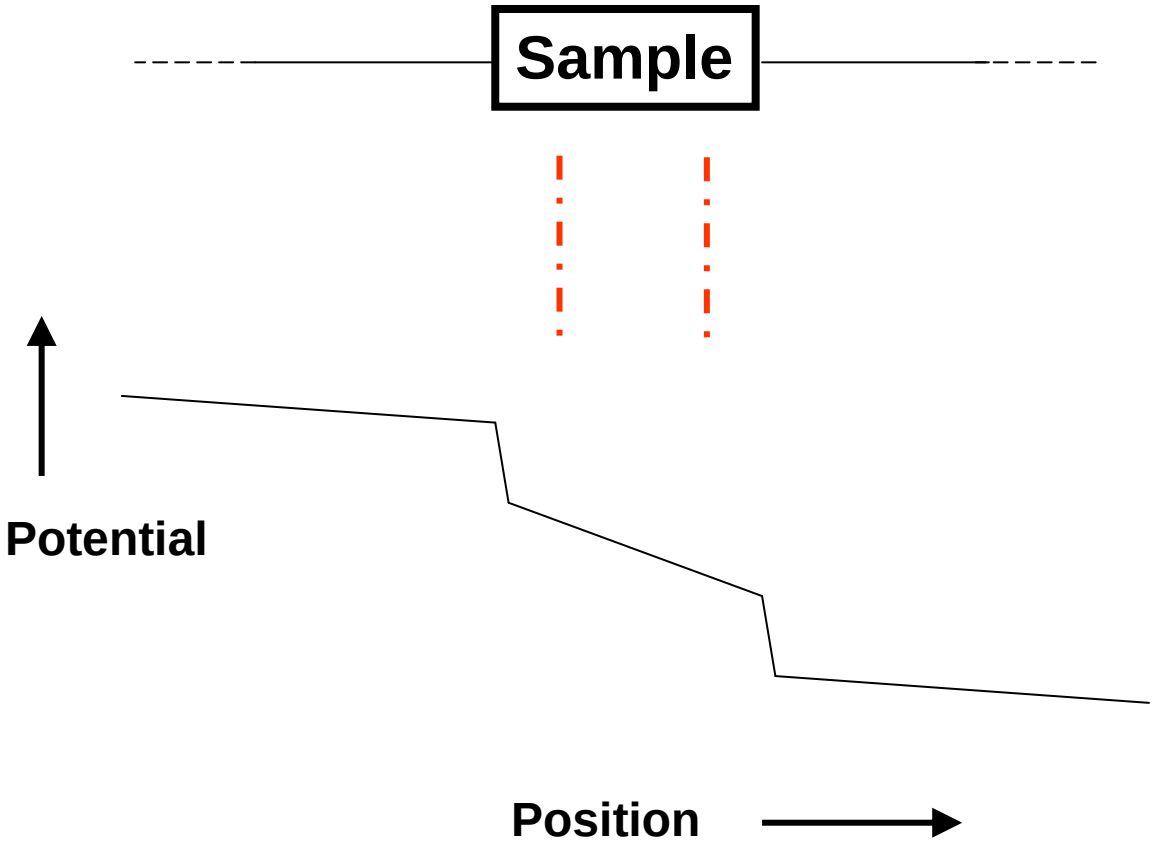


Figure 4.5: Potential as a function of position. The red dotted lines indicate the positions at which if the potentials are measured, then the potential drop measured can be more accurately attributed to the sample alone, and not to the contact resistance. This arrangement is used in the four probe measurement technique

We should also note, that no matter what we do, we may not be able to reduce R_c beyond a point. Therefore the trick to estimating V attributed to the sample more accurately, is to reduce the current going through the contact resistance, without reducing the current going through the sample. This may seem impossible at first glance, but is actually achieved quite easily since by nature of the functioning of a Voltmeter, only an extremely small current flows through the voltmeter. This fact is taken advantage of, in the four probe measurement technique. In this technique also the contact resistances cause a significant potential drop as seen from Figure 4.5 above. However, the voltmeter leads contact the sample in the region where only the sample defines the potential drop. There will still be a contact resistance associated with the contact between the voltmeter leads and the sample, however, the current flowing through the volt meter circuit will be several orders of magnitude smaller than 'I', the current flowing through the sample, and hence the errors caused are greatly reduced, and for all practical purposes virtually non existent.

The manner in which connections are made to enable the four probe conductivity measurement, are shown in Figure 4.6 below.

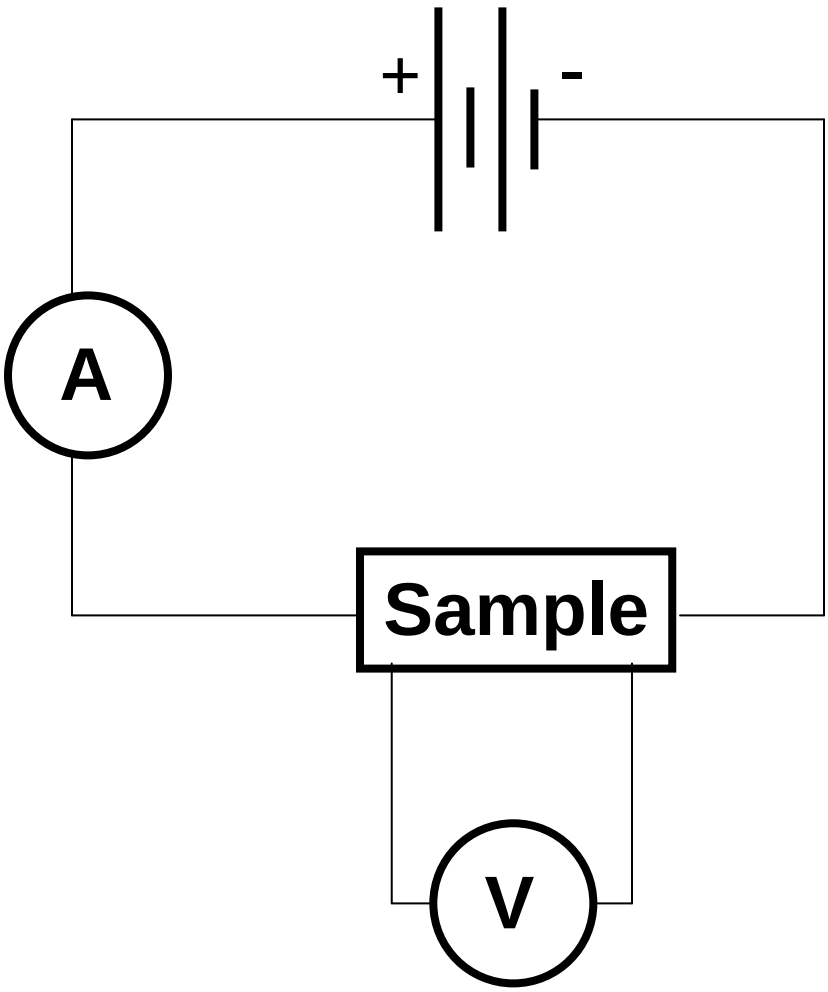


Figure 4.6: A four probe measurement of the electrical conductivity of a sample. A battery serves as the source of DC power, an ammeter measures the current in the circuit, and a voltmeter measures the potential drop across the sample. The voltmeter leads contact the sample within the region defined by the sample and hence do not measure the potential drops due to the contact resistances on either side of the sample.

Alternating Current (AC) Conductivity measurement:

We have seen two examples where the charge carriers are different in different locations in a circuit. Supposing we wish to measure the conductivity of a sample that uses ions as charge carriers, say for example oxygen ions, O^{2-}, the conductivity we measure is referred to as ionic conductivity. Ionic conductivity is different from the electronic conductivity that we are more commonly used to when we measure conductivity of wires. The issue we face when we try to measure ionic conductivity is that most of our measuring instruments such as voltmeters and ammeters work using electrons. These instruments are not designed to flow ions through them. Similarly several (but not all) ionic conductors

are designed to ensure that they have minimal to no electronic conductivity in view of the end use they are aimed for. Therefore a situation arises where we wish to determine the conductivity of a sample with a particular charge carrier, while the measuring instruments use a different charge carrier. When such a sample is connected in a typical DC conductivity measurement setup such as the one shown in Figure 4.6, a problem arises at each of the sample-circuit interfaces, where the wires from the external circuit contact the sample. While the DC power source tries to send electrons into the sample, the sample is unable to conduct the electrons, so there is a buildup of charge within the sample in response to the potential applied, in just the manner that a capacitor responds to the application of a potential across its terminals. The buildup of charge inside the sample, by the movement of oxygen ions, opposes the buildup of charge at the electrodes contacting the sample due to the movement of electrons in the external circuit. The charge buildup in the circuit is as shown in the Figure 4.7 below.

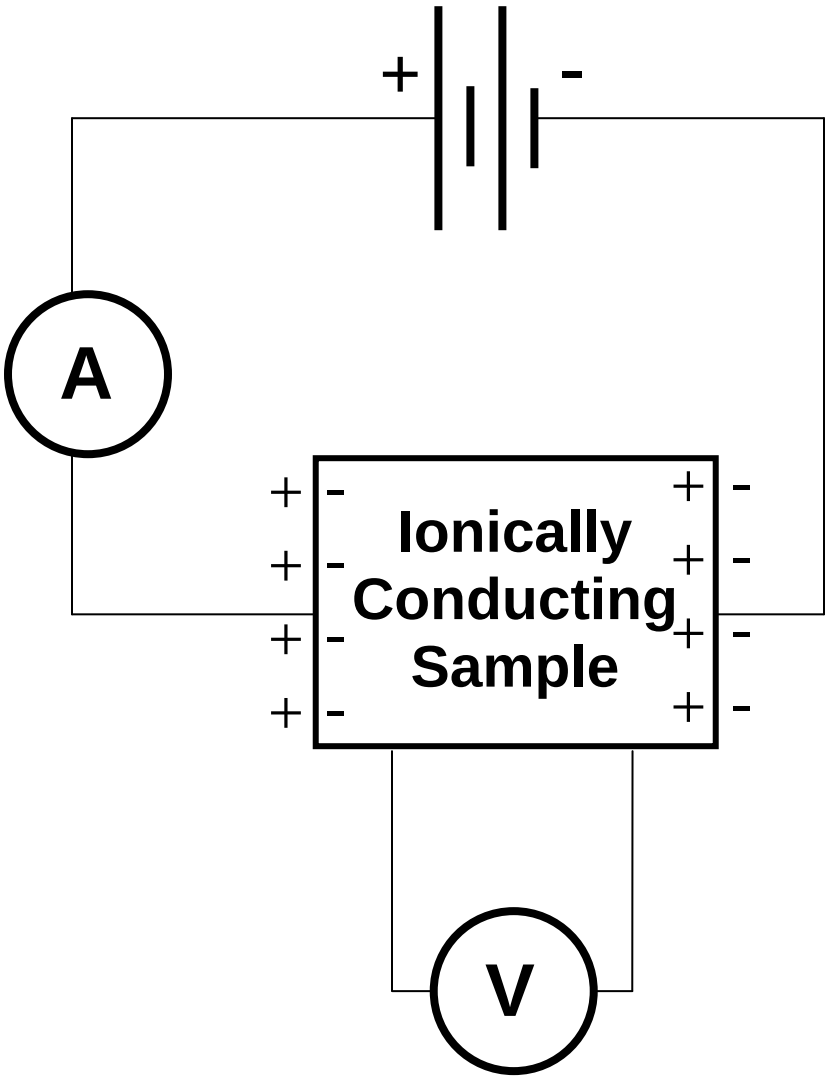


Figure 4.7: Buildup of charge when a sample which only conducts ions, is connected to a DC power source.

As a result of the charge buildup within the sample, the current in the circuit almost immediately drops to zero. Even during this brief interval, the current varies as a function of time, even though the voltage imposed is constant. Instruments with very high measuring speeds can measure the decay of current with time and this data can be analyzed to obtain information about the sample.

To measure conductivity in situations where there are different charge carriers, it is better to apply an alternating current, where the direction of current flow changes with time and hence the charge in the sample is also forced to swing back and forth within the sample. This type of measurement is referred to as AC conductivity measurement. AC conductivity can also be used for samples that conduct electrons, and hence is a more versatile technique when compared to the DC conductivity measurement. However, the equipment required to measure AC conductivity is typically much more expensive than DC measurement instruments and hence, where it is sufficient, DC conductivity is the preferred technique.

The schematic of the circuit used to carry out an AC conductivity measurement is shown in Figure 4.8 below and is similar to the one used for DC measurement, except that it uses an AC power source.

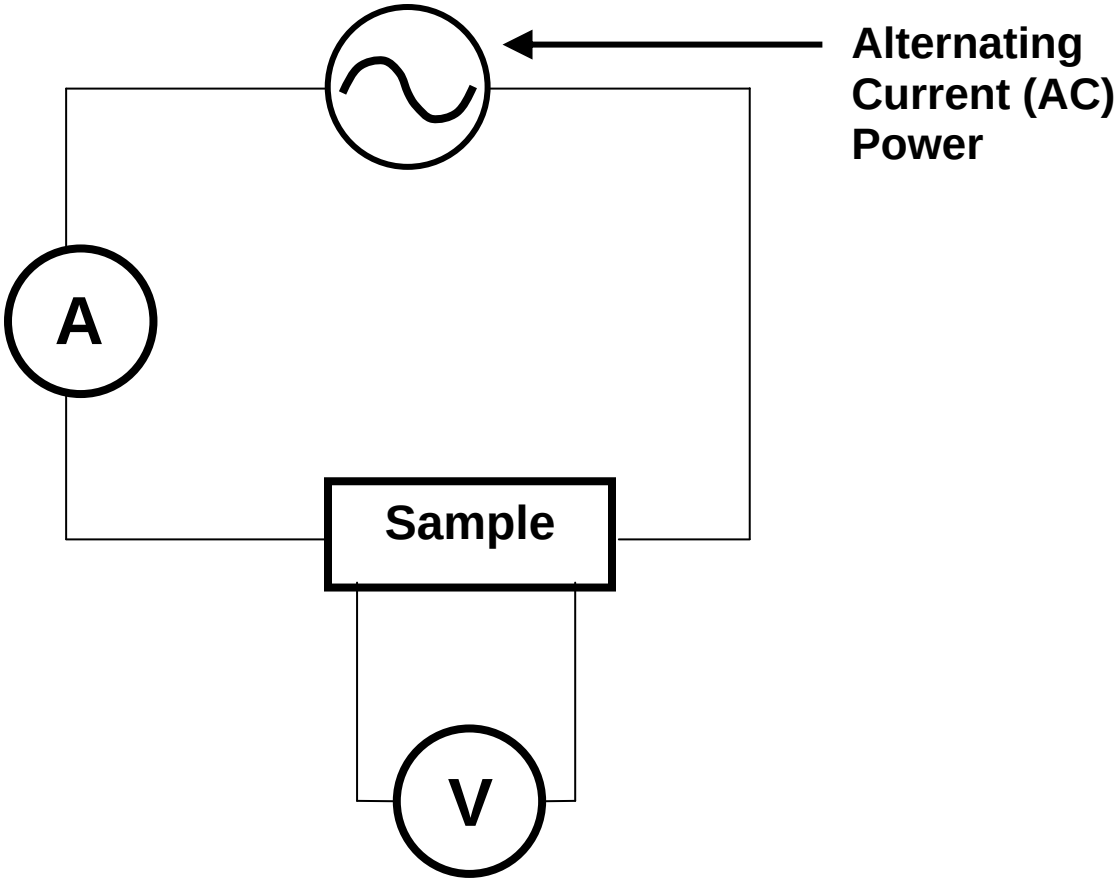


Figure 4.8: An AC four probe measurement of the conductivity of a sample.

In case the sample is a pure resistor, then Figure 4.9 below show schematically how current and voltage will vary with time in the circuit.

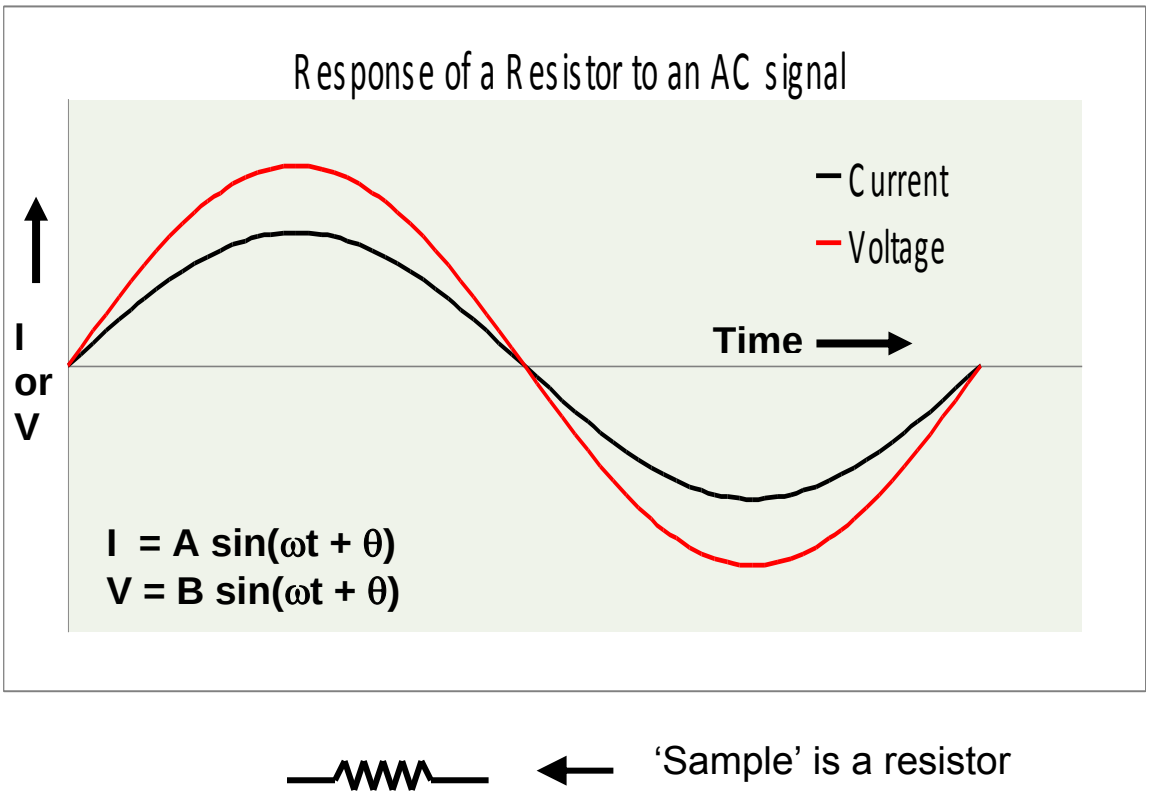


Figure 4.9: The response of a pure resistor to an applied AC signal.

In order to analyze data obtained using AC signals and determine the conductivity of the samples being tested, it is useful to understand some of the nomenclature and analysis techniques associated with AC signals.

AC signals vary with time and the current and voltage can be represented using equations of the form:

$I = A \sin(\omega t + \theta)$

$V = B \sin(\omega t + \theta)$

Where 'ω' is the angular frequency and is equal to $2\pi\upsilon$, where 'υ' is the frequency of the signal.

In a DC measurement, the measurement process involves measuring a single data point. In a typical AC measurement, the frequency of the AC signal is an experimental variable and is a valuable tool in probing the properties of the sample. The electricity supplied in homes typically has a fixed frequency of 50 Hz or 60 Hz around the world. However, this is just one of the possible frequencies that can be accomplished experimentally. In fact,

for carrying out AC conductivity measurements, typically a very wide range of frequencies is employed – from mHz to several kHz. The sample is examined at several frequencies from within this range, and considerable information can be obtained about the sample behavior and its fundamental properties in this process. Unlike the single point DC measurement, the AC measurement therefore involves recording several data points – one each at a series of specified frequencies. The sample is subject to a relatively small amplitude AC signal (say an AC voltage) and the instruments record the AC current response to this imposed signal. The amplitude of the signal used has to be large enough to enable a recordable response from the system, however it should also be small enough to ensure that the system responds linearly. During the experiment, variation in voltage as well as current, are recorded as a function of time. The ratio of these quantities, appropriately determined, as described below, indicates the response of the sample to the signals imposed on it.

While in the case of DC measurements on a pure resistor, we discuss the results in terms of resistivity, the more general phenomena as investigated using the AC technique, is referred to as 'Impedance' and is denoted by 'Z'. Impedance represents the tendency to obstruct the flow of current, and is the AC analogue of DC resistance. In the case of a pure resistor, 'Resistance' and 'Impedance' are exactly the same. In view of the 'impedance' offered to the flow of current, AC conductivity measurement technique is also referred to as the AC impedance technique.

To more conveniently analyze AC data, complex number notation is used. Let us briefly look at how an AC signal is denoted using complex number notation and also consider the validity of such a representation.

As shown in Figure 4.10 below, an AC signal can be thought of as having an 'x component' and a 'y component' at any given instant of time associated with the same modulus of the current 'I'.

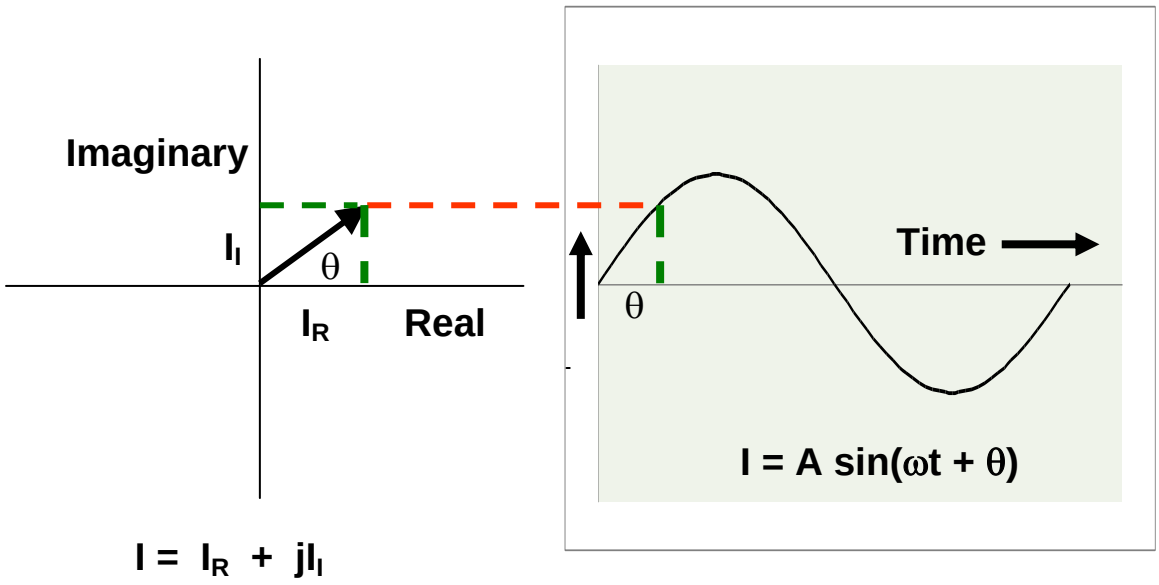
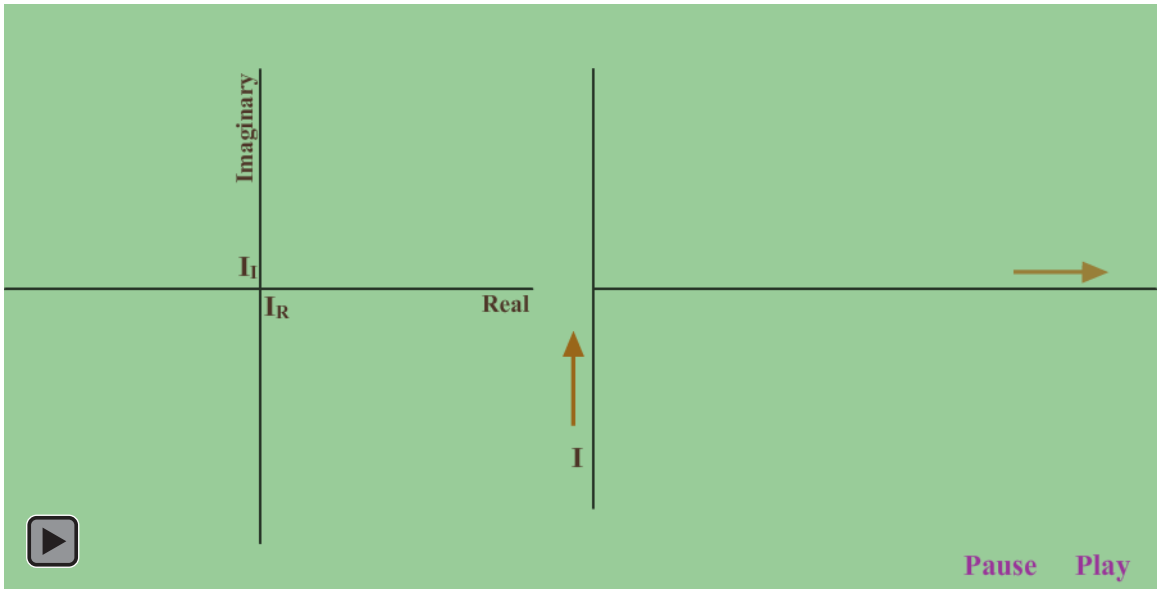


Figure 4.10: Representing an AC signal, in this case an AC current using complex number notation. Here 'j' is $\sqrt{-1}$, and I_r and I_i are the real and imaginary components of I, in the usual complex number notation

Animation of figure 4.10



The AC signal or waveform can be thought of as a vector of fixed modulus 'I' rotating about the origin such that the angle (and hence time) is obtained as $\tan^{-1}(I_i/I_r)$, and the amplitude $I = (I_i^2 + I_r^2)^{1/2}$. I_r and I_i are the real and imaginary components of I, in the usual complex number notation, and 'j' is $\sqrt{-1}$

The simplicity that complex number notation offers is that multiplying any vector by j, rotates the vector counter clockwise by 90°. Therefore, for example, multiplying a vector by j twice rotates it by 180° and hence makes it opposite to the original vector. Therefore by recognizing that the AC signal or waveform has varying 'x' and 'y' components associated with the same modulus of the current 'I', and by denoting the waveform using complex number notation, it is possible to capture the details of the current and voltage quantities accurately, and to use complex number mathematics to understand the interactions and implications of the quantities.

The response of a pure resistor to an AC signals, and hence the impedance offered by the resistor to the flow of current, can be identified using the AC impedance technique as shown in Figure 4.11 below.

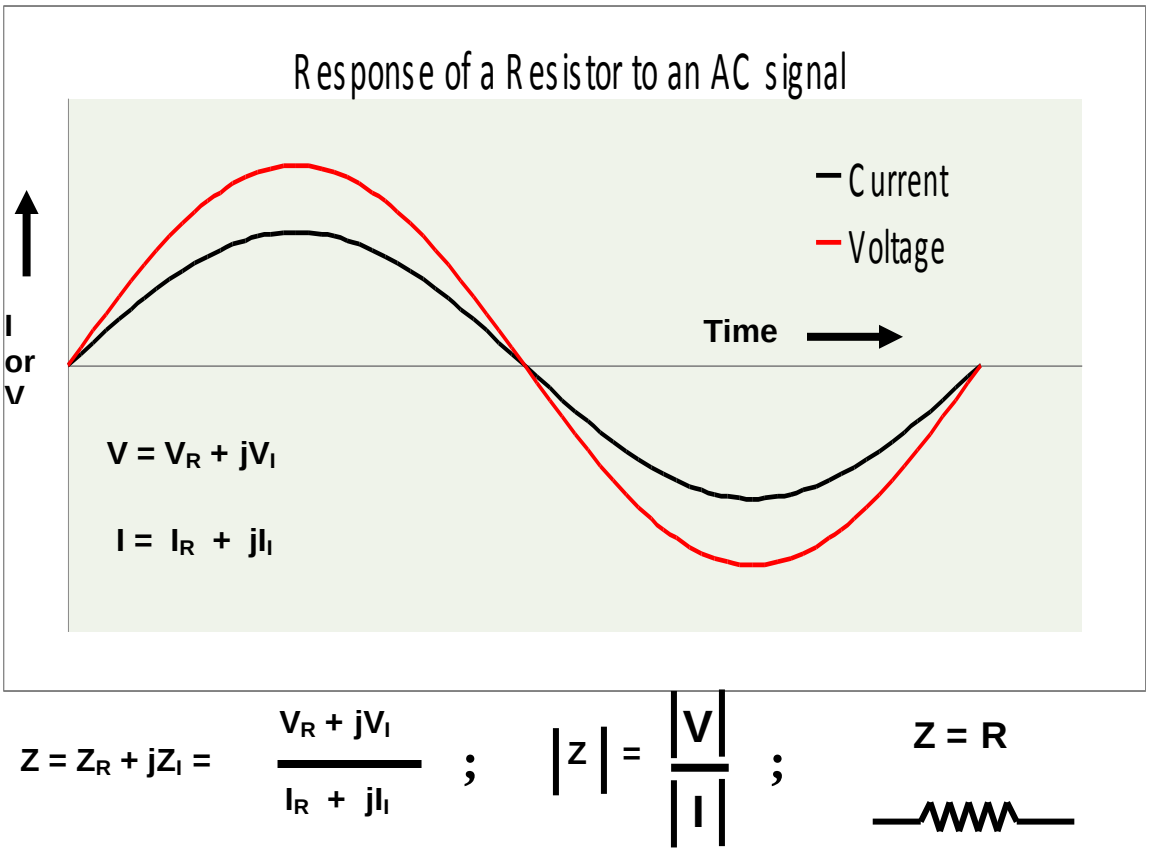


Figure 4.11: Response of a pure resistor to an AC signal. Current and voltage vary as a function of time but are exactly in phase. The impedance 'Z' is equal to the DC resistance 'R'.

In the case of a pure resistor, the impedance 'Z' is not a function of the frequency. It is equal to the resistance 'R' regardless of the frequency used to make the measurement.

When the sample is a pure capacitor, current and voltage are not in phase. Current leads voltage by 90°. This phase difference results in the impedance offered by a capacitor being a complex quantity.

Figure 4.12 below shows the response of a pure capacitor to an AC signal.

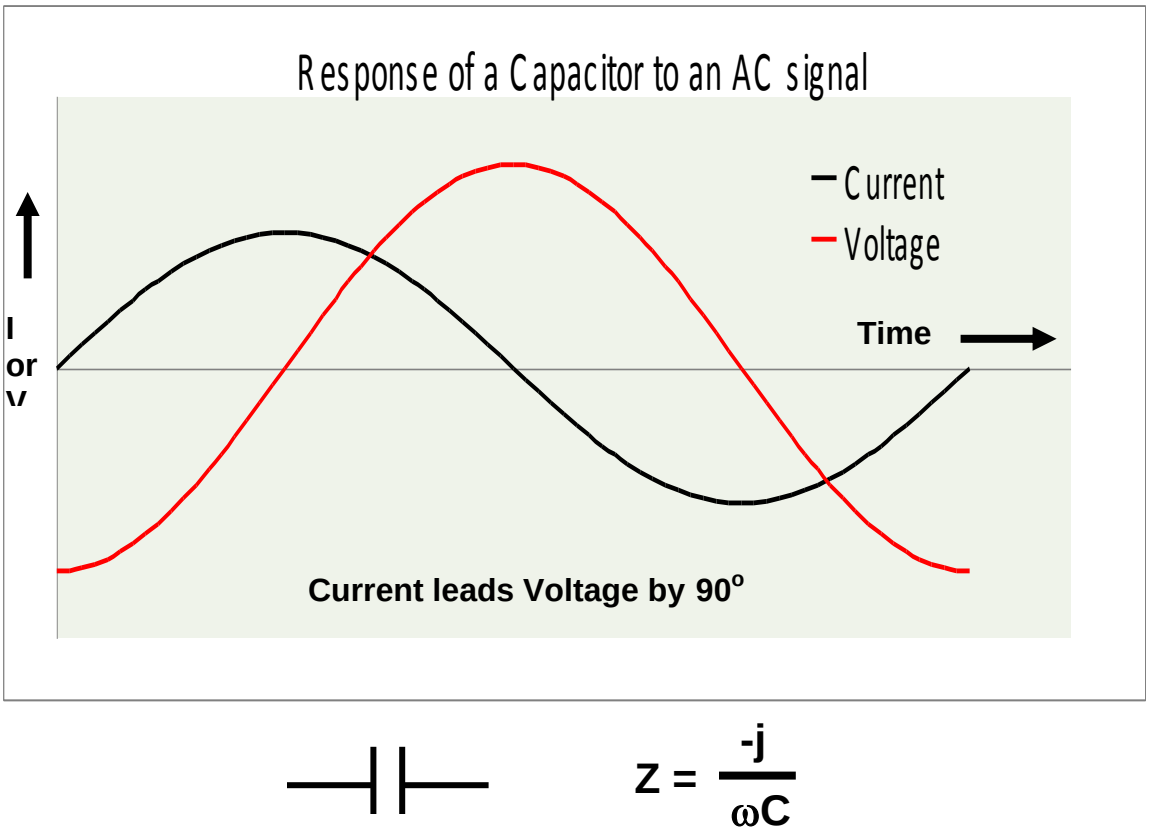
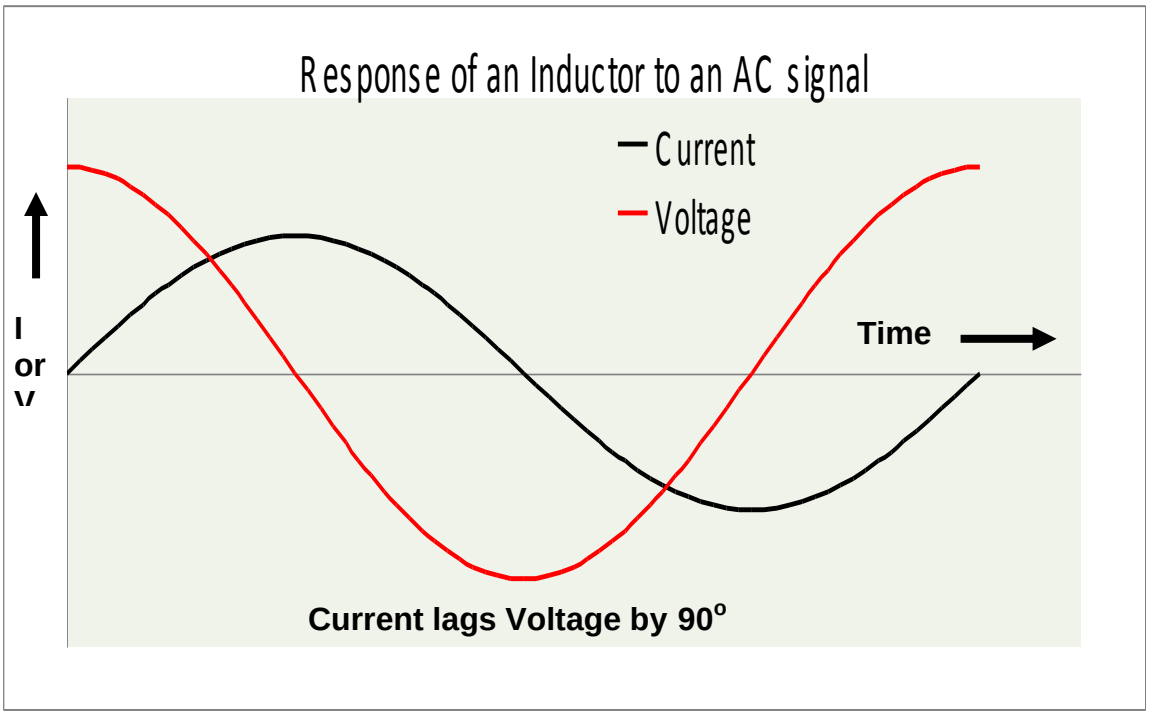


Figure 4.12: Response of a pure capacitor, of capacitance C, to an AC signal. Current and voltage vary as a function of time current leads voltage by 90°. The impedance 'Z' is a complex imaginary quantity and is a function of the frequency ω.

The impedance of a capacitor is seen to depend on the frequency used to make the measurement. At very high frequencies (ω is high), the impedance 'Z' drops to zero and the capacitor behaves as though it has been shorted internally. At very low frequencies, Z becomes a very high value and becomes infinity when ω drops to zero - or when a DC signal is employed, which is consistent with the behavior of a capacitor in a DC circuit.

When an inductor is subject to an AC signal, its behavior is as shown in Figure 4.13 below:



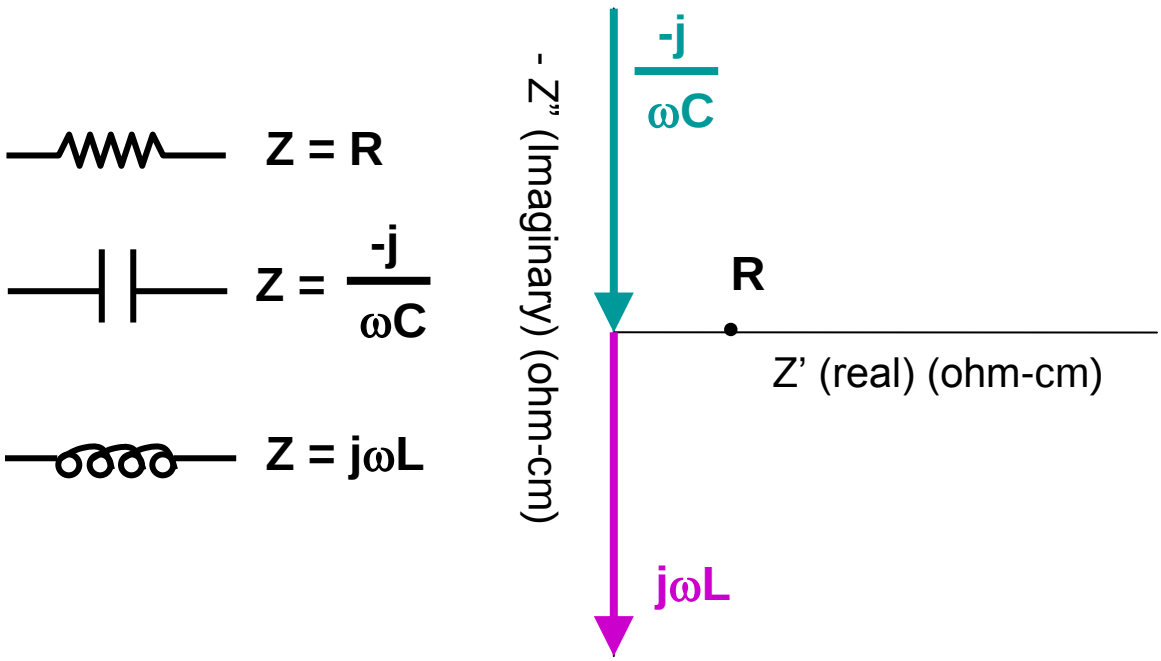


$Z = j\omega L$

Figure 4.13: Response of a pure inductor, of inductance L, to an AC signal. Current and voltage vary as a function of time and current lags voltage by 90°. The impedance 'Z' is a complex imaginary quantity and is a function of the frequency ω.

The inductor shows a behavior that is descriptively the inverse of the behavior shown by a capacitor, when subject to an AC signal. At high frequencies its impedance is high, whereas it behaves as though it is internally shorted when the frequency of the AC signal drops to zero or, in other words, when a DC signal is used.

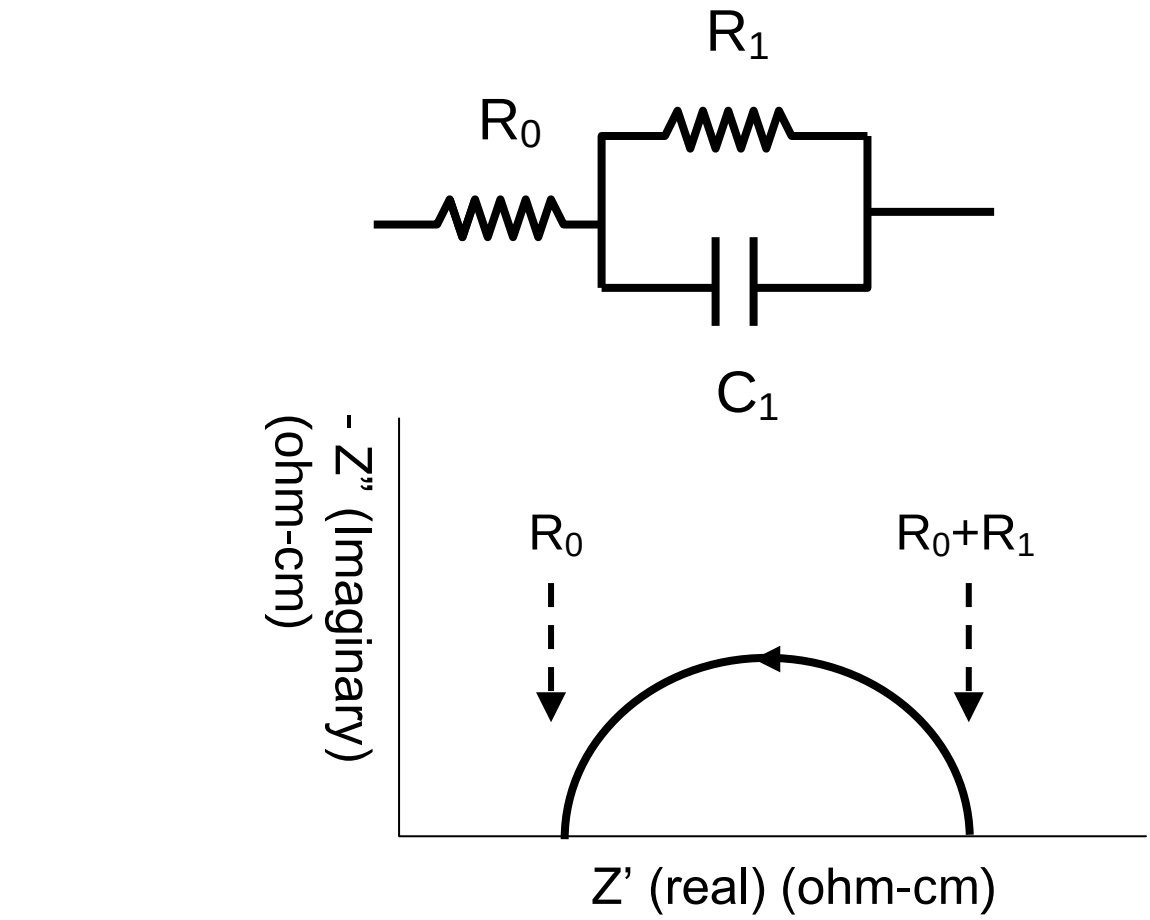
The response of the three circuit elements discussed so far, a resistor, a capacitor, and an inductor, as a function frequency of the AC signal used, is summarized in Figure 4.14 below. In each case, the data consists of a series of points, one each at specific frequencies, measured over a range of frequencies. In the case of a resistor, the points measured coincide within experimental error. In the case of capacitors and inductors, the points measured coincide with the y axis. Please note, the negative of the imaginary impedance is plotted on the y axis as a matter of convention since in many of the systems investigated capacitive responses are prominent.



Click here for animation on
Impedance as a function of increasing frequency for R, C, L

Figure 4.14: Impedance of a pure resistor R, a pure capacitor C and a pure inductor L to an AC signal. The arrows indicate the direction of increasing frequency ω. The impedance of the resistor is unaffected by the value of ω. Z' is Z_r, and Z" is Z_i. As a matter of convention, -Z" is plotted on the y axis.

While the discussion so far has looked at individual circuit elements such as a pure resistor, or a pure capacitor, real systems display characteristics that are equivalent to having a combination of resistors and capacitors. Figure 4.15 below shows a possible combination of resistors and capacitors and the resultant impedance behavior as a function of frequency.



$$\omega_{\text{maximum imaginary}} = \frac{1}{R_1 C_1}$$

Animation of the above is shown in the next page

Figure 4.15: Impedance of a circuit containing pure resistors R_0 and R_1, and a pure capacitor C_1 The solid arrow indicates the direction of increasing frequency ω. The dotted arrows indicate the intercepts.

At high frequencies the capacitor behaves as if it is internally shorted, therefore the impedance is only R_0. At very low frequencies, the impedance of C_1 is almost infinite and hence the current flows through R_0 as well as R_1 and the impedance is $R_0 + R_1$. At intermediate frequencies, the impedances trace a semicircle as shown in Figure 4.15.

AC impedance analysis is used to study complex systems where several phenomena may be occurring in series or in parallel. By subjecting the system to an AC signal the phenomena are forced to oscillate back and forth at each of the specific frequencies. The ease or difficulty with which the phenomena are able to follow the AC perturbation then decides the response of the system.

Animation of figure 4.15: RC Circuit



AC impedance data are analyzed by a few different approaches. In its simplest form the AC impedance data are obtained from the sample or system under various experimental conditions or after the sample has been subject to various experimental conditions, and the data are compared. Features such as the location of specific intercepts, the size of any semicircles observed in the data etc, are noted and inferences are made on what has happened to the system based on prior experience with such systems.

A more sophisticated approach requires theoretically fitting the data to an equivalent circuit. Each element of the circuit is then associated with a physical phenomenon in the sample and the changes in the value of the circuit element with experimentation is interpreted as changes in the parameters associated with the phenomenon. It is important to note that several circuits can simulate the same data. Therefore the choice of circuit can greatly impact the effectiveness of the interpretation. Considerable experience is required to utilize this approach successfully.

An even more rigorous manner to handle AC impedance data is to compare it with theoretical models of the system. In this case the theoretical models already incorporate the fundamental phenomena involved, and therefore when the theoretical curve matches the experimental data, interpretation is much easier than in the case of equivalent circuit fitting.

In the curve fitting discussed above, one additional aspect is important and different from that in typical data fitting encountered in engineering and science. In AC impedance analysis, each data point is obtained at a specific frequency. Even in the simulated data, each data point corresponds to a specific frequency. For a good fit, it is important that at each frequency the experimental and theoretical data points match. In other words, for example, the data point experimentally obtained at 35 Hz should match that obtained theoretically at 35 Hz. The fit is not considered acceptable if only the shape and size of the curves match, while the specific data points themselves do not match.

Short note on Superconductivity:

Superconductivity is a phenomenon that is displayed by some materials at very low temperatures. The present understanding of this phenomenon relates it significantly with magnetism and indicates a mechanism that is quite different from that displayed by materials at room temperature. Superconductivity is therefore described in greater detail in a separate class, later in this course.